



Bird community responses to habitat fragmentation: how consistent are they across landscapes?

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ABSTRACT

Aim The woodland ecosystems of south-eastern Australia have been extensively disturbed by agriculture and urbanization. Herein, the occurrence of birds in woodland remnants in three distinct landscapes was analysed to examine the effects of different types of landscape matrices on species richness vs. area and species richness vs. isolation relationships and individual species responses to woodland fragmentation.

Location The study system comprised three distinct woodland landscapes of the northern Australian Capital Territory and bordering areas of New South Wales. These landscapes (termed agricultural, peri-urban and urban) are located within 50 km of each other, have remnant fragments of similar age, size, isolation, woodland cover, elevation and climates. The major distinguishing feature of the three landscapes was the properties of the habitats surrounding the numerous woodland remnants.

Methods Bird surveys, using an area-search methodology, were conducted in 1999 and 2000 in 127 remnants in the three landscapes to determine bird species presence/absence. Each remnant was characterized by measures of remnant area, isolation and habitat complexity. To characterize differences between each landscape, we conducted an analysis of the amount of tree cover and human disturbance in each landscape using SPOT imagery and aerial photographs. Linear regressions of woodland-dependent species richness vs. remnant area and remnant isolation for the three different landscapes were calculated to see if there were any apparent differences. Binomial logistic regressions were used to determine the relationships between the occurrence of each species and the size and isolation of woodland habitat, in each landscape.

Results All the landscapes displayed a significant ($P < 0.01$) species vs. area relationship, but the slope of the urban relationship was significantly greater than those of the other landscapes. In contrast, only the agricultural landscape displayed a significant ($P < 0.01$) species richness vs. isolation relationship. When individual species were investigated, we found species that were: (1) apparently insensitive to reduction in remnant area and increase in isolation across all landscapes, (2) absent in small remnants in all landscapes, (3) absent in small remnants in all landscapes and also absent in isolated remnants in the agricultural landscape, (4) absent in isolated remnants in the agricultural landscape, and (5) absent in small remnants in the urban landscape. Threshold values (50% probability of occurrence) for area and isolation for individual species were highly variable across the three landscapes.

Main conclusions These results indicate that woodland bird communities have a varying response to habitat fragmentation in different landscapes. Whilst we cannot be sure how representative our chosen landscapes are of other similarly

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composed landscapes, these results suggest that the type of landscape matrix may have a considerable influence on how bird species are affected by woodland fragmentation in the region. For instance, the properties of a matrix may influence both the resources available in the landscape as a whole for different bird species, and the connectivity (dispersal of birds), between woodland remnants. We encourage further research that examines these hypotheses and argue that the management of the matrix should be included in conservation strategies for fragmented landscapes.

Keywords

Australia, birds, fragmentation, island biogeography, matrix effect, species–area effect, species–isolation effect, woodland.

INTRODUCTION

Human activities such as forest clearance and the expansion of agricultural land have exacerbated the natural fragmentation of landscapes (Andr  n, 1994). In the past two decades there has been increasing interest in studying the effects of anthropogenic habitat fragmentation on the spatial dynamics of populations and communities. In particular, the effects of reducing the size of habitat patches (area effects), increasing the distance between occupied patches (dispersal effects), and the creation of habitat edges on patches (edge effects) have dominated the conservation literature (reviewed in Saunders *et al.*, 1991; Turner, 1996; Debinski & Holt, 2000). Many studies have demonstrated that these three processes have been detrimental to indigenous biodiversity throughout the world.

One important aspect of habitat fragmentation, the creation of a different (often human dominated) matrix that surrounds habitat fragments and its impacts on animal communities that inhabit habitat fragments, has been less well studied (Lindenmayer & Franklin, 2002 but see   berg *et al.*, 1995; Malcolm, 1997;   s, 1999; Brotons *et al.*, 2003). We suggest that the role of the matrix has been given insufficient attention, perhaps as a result of the dominance of MacArthur & Wilson's (1967) equilibrium theory of island biogeography, and the related concepts of rescue effects (Brown & Kodric-Brown, 1977), nestedness (Patterson & Atmar, 1986), and metapopulation dynamics (Levins, 1968; Hanski & Gilpin, 1991), as theoretical frameworks in many fragmentation-related studies (Lomolino & Smith, 2003). This body of work is based on the concept of populations inhabiting suitable habitat patches and being surrounded by non-suitable habitat (Doak & Mills, 1994; Whittaker, 1998; Ricketts, 2001).

In the present paper, we present a within-region analysis of three bird data sets, all collected recently in different fragmented woodland landscapes in and around Canberra, in south-eastern Australia. Each of the three data sets was taken from 'variegated' (sensu McIntyre & Hobbs, 1999) landscapes that have distinctly different landscape contexts (i.e. different matrices). We defined these landscapes as: 'agricultural', 'peri-urban' and 'urban'. The landscapes are

otherwise very similar, as modern clearance and modification occurred approximately at the start of the twentieth century, and each landscape has analogous remnant size and spatial pattern distributions (Freudenberger, 2001; Watson *et al.*, 2001). We show below that the major difference in the landscapes appears to be in the properties of the landscape matrix surrounding the woodland remnants. For this reason, and because of the close proximity of the landscapes to each other (thus negating a climate effect), we were able to assess how woodland bird communities respond to habitat fragmentation in landscapes of substantially different matrix compositions.

Our first aim was to determine if bird species richness–area and species richness–isolation relationships differed in fragmented woodland remnants in the three landscapes. We hypothesized that there would be different species richness–area and species richness–isolation relationships between each of the three landscapes, for a number of reasons. First, the 'edge-contrast' between remnant and matrix has been shown to be an important predictor for bird diversity within fragmented landscapes (Watson *et al.*, 2004). Edge-contrast governs the micro-climate gradient at the edge of a remnant, which in turn could lead to substantial vegetation changes including changes in floristic composition and structure (Kapos *et al.*, 1997; Rankin-De M  rona & Hutchings, 2001). Edge effects can also cause increases in predators, nest predators and nest parasites along the new edge (Robinson *et al.*, 1995). Secondly, different matrices have different resources within them, and therefore have differential capacity for different species – if a species cannot use the resources of a particular matrix, remnant patches may have to be larger to support the species (Cook *et al.*, 2002; Lindenmayer *et al.*, 2003). Thirdly, the contrast between fragments and matrix may determine the degree of isolation a local population experiences, because the properties of a landscape matrix may affect the ability of individuals to move through it, which will influence population persistence in fragmented landscapes (Stouffer & Bierregaard, 1995a,b; Sisk *et al.*, 1997; Gascon *et al.*, 1999; Renjifo, 1999; Wethered & Lawes, 2003). All these factors affect the distribution of species in fragmented

landscapes, and all may potentially change if matrix composition differs.

Our second aim was to determine which species were sensitive to area loss and patch isolation in each landscape, and whether they were affected in the same way in all three landscapes. We did this by taking a species-level approach and classifying the species response to the five biologically realistic outcomes for how a species may react to remnant area and isolation (Lomolino, 1984, 1986; Lawlor, 1998; Watson, 2003). These are: random, minimum area, maximum isolation, non-compensatory area and isolation and compensatory patterns (Fig. 1). We investigated if these patterns of distribution were consistent for individual species in all three landscapes.

MATERIALS AND METHODS

Study area

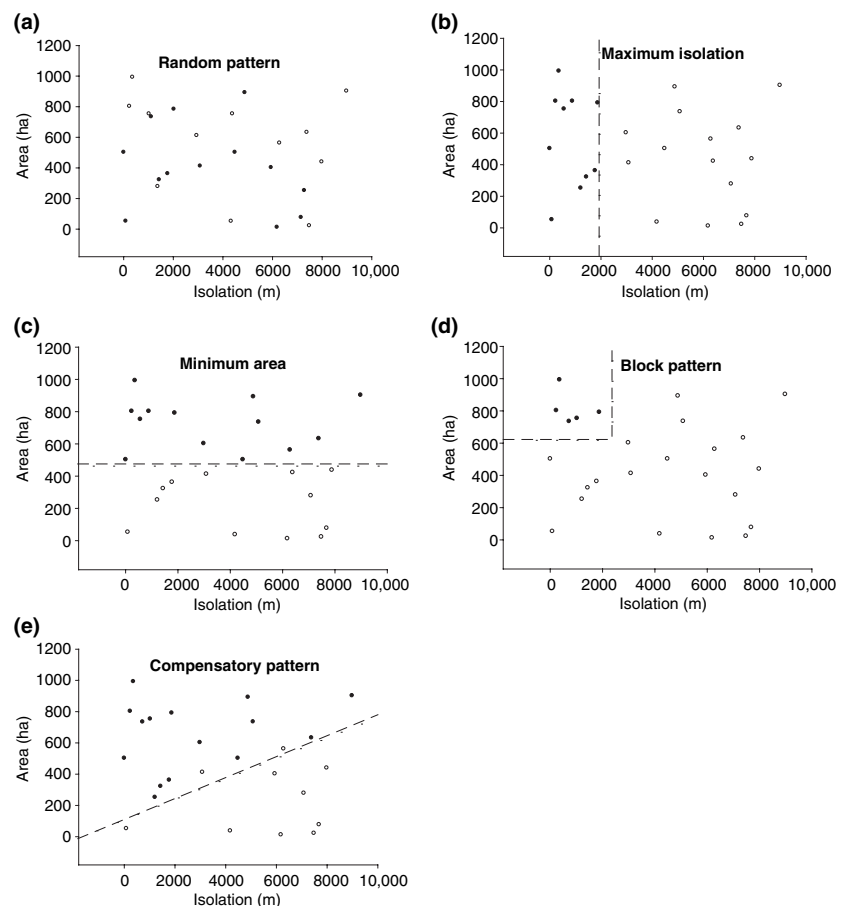
The study area included the northern part of the Australian Capital Territory (ACT) and bordering areas of New South Wales, and consisted of undulating hills and floodplains surrounding the Murrumbidgee, Molonglo and Boorowa River catchments (Fig. 2). Before European colonization in the early nineteenth century, grasslands and woodlands comprised the dominant vegetation communities (Burbidge & Gray, 1970). Since then, these vegetation communities have been largely

cleared of native vegetation due to the development of the city of Canberra and agricultural activities including cultivation for crops and introduction of exotic pasture grasses and legumes. The temperate woodlands in this region are now one of the most fragmented ecosystem types in Australia (Garnett & Cowley, 2000).

The 'agricultural' woodland landscape comprised the Boorowa River catchment, which covers approximately 168,000 ha on the south-west slopes of New South Wales. Agriculture, including wool, beef, cereal and some horticultural production, is the dominant land-use. The catchment has been extensively cleared and only about 7% of it remains under woodland cover in patches at least 1 ha in size. Numerous outbreaks of dryland salinity have been mapped within the catchment, with 1372 ha of salt-affected areas identified within an 83,000 ha mapped area in 1995 (Hayman, 1996).

The 'peri-urban' landscape comprises 150,000 ha of the northern ACT and bordering areas of New South Wales, and consists of undulating hills and floodplains of the Murrumbidgee and Molonglo rivers. This landscape is not characterized by intense cultivation and annual crops, but instead has been modified by the establishment of introduced pastures, grazing by livestock, urbanization and selective clearing (e.g. firewood removal). The establishment of small (< 50 ha) 'hobby' farms over the past 50 years has been the dominant form of urbanization in this landscape. It still has

Figure 1 Hypothetical distributions of species, arranged relative to fragment area and isolation (after Lomolino, 1986; Lawlor, 1998). Open circles are unoccupied fragments, while closed circles represent fragments in which the species occurs. (a) Random pattern in which the distribution of species is not affected by area or isolation. (b) Maximum isolation pattern in which species are dispersal-limited, and recorded only in those fragments below a threshold level of isolation. (c) Minimum area pattern in which species are area-sensitive and inhabit only those fragments larger than a particular threshold. Two patterns may be exhibited by those species whose insular distributions depend on both area and isolation. (d) A 'block' pattern occurs when a species depends on resources of the largest remnants and has limited ability to immigrate. (e) A 'compensation' pattern in which isolated fragments are occupied when of sufficient large size, and small fragments are occupied when close to larger patches.



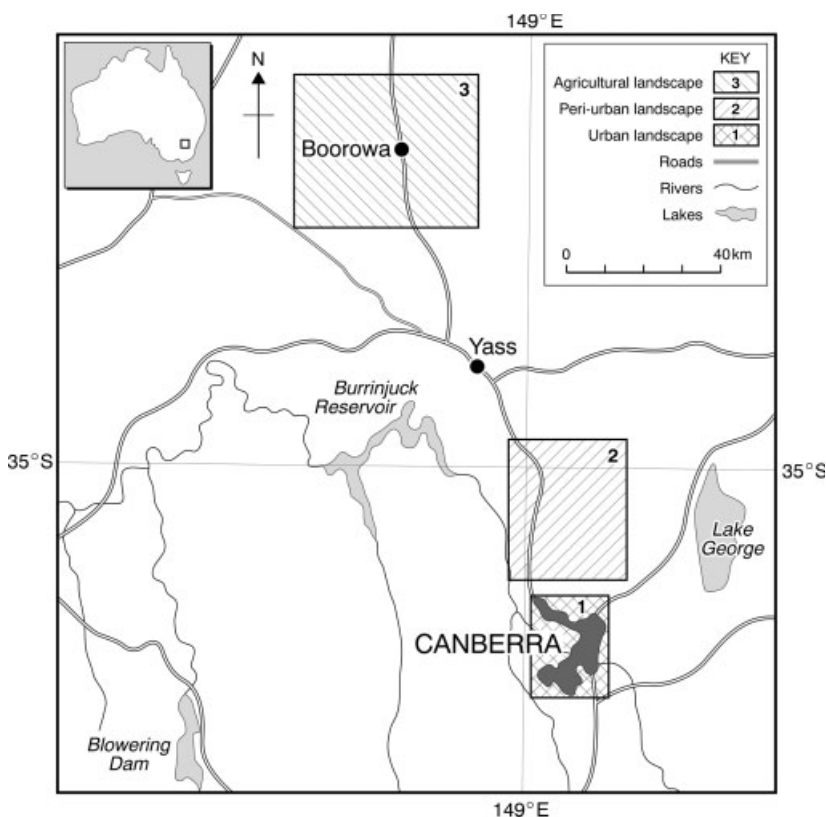


Figure 2 The location of the three woodland landscapes (agricultural, peri-urban and urban) surveyed in south-eastern Australia.

about double the number of scattered trees in fields compared with the agricultural landscape of the Boorowa catchment, but many more roads and houses.

The 'urban' landscape is within the realms of the city of Canberra, the major urban centre within the region (population 313 000). This landscape covers approximately 150,000 ha. Within the city, woodland patches remain as parks and hillside reserves surrounded by the urban matrix, which is dominated by housing, roads and a diversity of native and exotic vegetation in gardens and along suburban streets.

The floristic composition of woodland remnants in all three landscapes is very similar. Associations of Yellow Box *Eucalyptus melliodora* Schauer and Blakely's Red Gum *E. blakelyi* Maiden dominate the tree cover of many remnants on lower slopes and deeper soils, while Red Box *E. polyanthemos* Schauer, White Box *E. albens* Miquel, Apple Box *E. bridgesiana* Baker, Mealy Bundy *E. nortonii* Blakely, and Ribbon Gum *E. viminalis* Hook are less common. The drier and steeper slopes are dominated by Red Stringybark *E. macrorhyncha* Benth, Brittle Gum *E. mannifera* Mudie, and Scribbly Gum *E. sclerophylla* Blakely. The shrub strata are typically composed of *Bursaria spinosa* Cavanilles, *Cassinia* spp. and *Acacia* spp. The ground layer in relatively undisturbed remnants is dominated by species such as Kangaroo Grass *Themeda triandra* Forsskal and Spear Grass *Stipa* spp. Degraded woodland remnants contain exotic understorey plants, including blackberries *Rubus* spp., roses *Rosa* spp. and introduced grasses (Watson, 1999; Freudenberger, 2001).

We chose these three landscapes not only for their different landscape matrix composition but also because they were in close proximity to each other, and as such, experience very similar climatic conditions (Bureau of Meteorology, 2004). Moreover, each landscape had similar remnant size, spatial pattern distribution and similar proportions of vegetation cover in each landscape (Table 1; Figs 3 and 4). The woodland cover ranged from 7% to 10%, well within the range where fragmentation effects are postulated to occur (Andr n, 1994). The extensive clearance and modification of all three landscapes by Europeans started approximately at the start of the twentieth century and the woodland remnants have similar ages since fragmentation.

Field methods

Bird presence/absence data were collected in 127 woodland remnants of different sizes in the three landscapes. Bird surveys in the urban ($n = 30$) and peri-urban ($n = 42$) landscapes were conducted during March–May 1999 and bird surveys in the rural landscape ($n = 55$) were conducted in March 2000. For the two year period prior to the bird surveys and during the bird surveys (March 1999–March 2000), the three landscapes had very similar climatic conditions, varying at most by 2 °C in temperature ranges per month throughout this period, and had a very similar rainfall regime. In addition, the three landscapes did not experience severe drought or flooding events during the period 1997–2000 that could have affected

Table 1 Summary statistics for woodland remnants and of matrix variables in each of the three landscapes (mean $\pm$ SE)

	Agricultural	Peri-urban	Urban
Remnant area range (ha)			
Area (min)	1	1	3
Area (max)	892	1616	571
Area (mean)	93	125	181
Area (median)	21	14	22
Remnant isolation range (m)			
Isolation (min)	10	68	40
Isolation (max)	6440	8700	5300
Isolation (mean)	1488	2470	1362
Isolation (median)	870	1271	802
Matrix measures			
Number of houses km ⁻²	0.12 (0.23)	0.31 (0.36)	701.2 (73.1)
Number of roads km ⁻²	0.4 (0.42)	0.98 (0.62)	48.9 (9.6)
Length (km) of roads km ⁻²	0.38 (0.42)	0.62 (0.43)	8.8 (1.5)
Number of trees km ⁻²	14.8 (5.6)	30.2 (14.2)	417 (162.3)
Number of clumps km ⁻²	0.54 (0.41)	0.35 (0.31)	77.4 (60.1)
Number of vegetated creeks km ⁻²	0.28 (0.31)	0.1 (0.15)	0.11 (0.4)
Length (m) of vegetated creeks km ⁻²	0.12 (0.15)	0.03 (0.04)	0.18 (0.85)

All measurements were conducted using the SPOT imagery in the peri-urban and agricultural landscapes. Aerial photographs captured in May 2000 at a spatial resolution of 1 : 20,000 were used for the urban landscape. All area measurements are in hectares (ha) and all isolation measurements are in metres. Remnant isolation was defined as the distance to the nearest remnant > 10 ha in size. Landscape matrix variables are means of measurements from 10 randomly placed 1-km radius circles in each landscape.

the total population size of a bird species in one or more of the landscapes (cf. Hinsley *et al.*, 1995, 1996).

In the agricultural landscape, woodland remnants were surveyed using three area-search 30-min surveys by two experienced field ornithologists (Bibby *et al.*, 1992). In these surveys, species occurrence and relative abundance were recorded. In the other two landscapes, woodland remnants were surveyed using three 20-min surveys by two experienced ornithologists. In these surveys, only species occurrence data were recorded. Species accumulation curves and pilot studies showed that the sampling effort was sufficient to capture accurately bird presence in every remnant. Full details of the census methodology and results of pilot studies are given in Watson (1999) and Freudenberger (2001). Because missing species that are actually present in a patch can distort species-incidence analyses if the errors are related systematically to patch size, we took the further step of comparing our results to studies conducted by the Canberra Ornithologists Group (COG) in 18 large (> 20 ha) woodland remnants in the ACT (Taylor & Canberra Ornithologists Group, 1992; Barrett *et al.*, 2003). We found the same species presence/absence that the COG surveys found, providing further support for the effectiveness and lack of bias of our survey. This makes it unlikely that the differences in species incidence functions between the different landscapes are distorted by either survey bias or our sampling the systems at particular points in a temporal dynamic responding to climatic stress.

Patch characteristics

A rapid appraisal procedure was used to measure the structural complexity of vegetation in the 127 remnants surveyed for

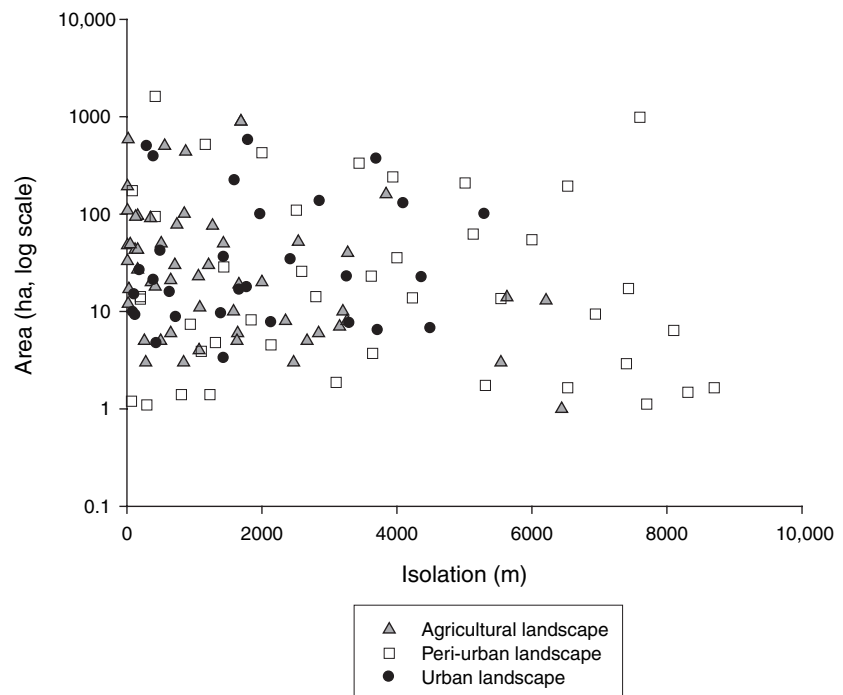


Figure 3 The distribution of surveyed remnants in terms of their size and isolation in agricultural, peri-urban and urban landscapes in south-eastern Australia. Remnant area and isolation were independent of each other (urban: $F_{1,28} = 0.26$, $P = 0.61$, $r^2 = 0.01$; peri-urban: $F_{1,40} = 0.48$, $P = 0.4$, $r^2 = 0.01$; agricultural: $F_{1,53} = 0.73$, $P = 0.39$, $r^2 = 0.02$).

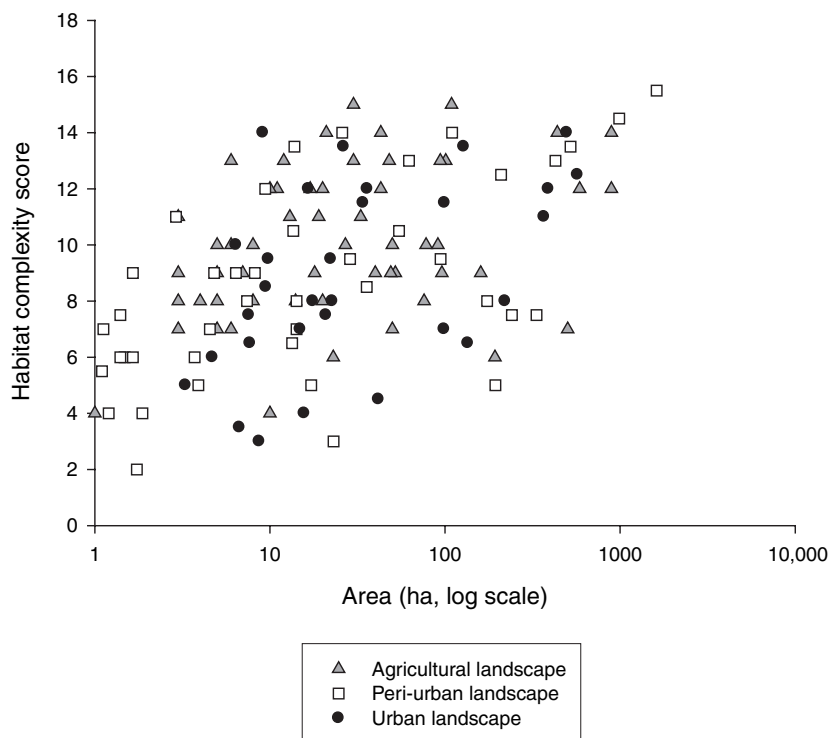


Figure 4 Area (A) and habitat complexity scores (HCS) of the surveyed remnants in the agricultural, peri-urban and urban landscapes. Remnant area and habitat complexity were correlated with each other [urban: $F_{1,28} = 7.1$, $P < 0.01$, $r^2 = 0.21$, $HCS = 8.1 + 1.41 \log(A)$; peri-urban: $F_{1,40} = 14.5$, $P < 0.01$, $r^2 = 0.27$, $HCS = 5.8 + 2.2 \log(A)$; agricultural: $F_{1,53} = 3.12$, $P < 0.01$, $r^2 = 0.12$, $HCS = 5.2 + 2.4 \log(A)$].

birds. Habitat complexity was recorded using randomly placed quadrats of 100 m × 100 m. After conducting a pilot study using habitat complexity accumulation curves per quadrat, it was found that two quadrats were required to represent variation in habitat heterogeneity in remnants sized between 2 and 10 ha, three quadrats for remnants of 10–50 ha, four quadrats for remnants of 50–100 ha and five quadrats for remnants > 100 ha (Watson, 1999). For each remnant, a mean habitat complexity score (HCS), modified from that described by Catling & Burt (1995), was derived on the basis of six attributes (Table 2). Resources relevant to birds and encompassed by these attributes included: nectivorous food supply; nesting sites and shelter from predators in the shrub layer; grass seeds in the ground layer; and invertebrate food in the litter layer. Each habitat attribute was rated on a scale of 0–3, and the scores for the six attributes were summed to calculate a HCS for each quadrat. Scores < 5 were typically obtained from woodland remnants with sparse shrub and ground layers, few logs and fallen branches, and not much litter. An HCS > 10 described structurally complex woodlands in which most habitat components were given a medium to high score.

The area of each remnant woodland patch was calculated after screen-digitizing their outlines from a SPOT panchromatic satellite image captured on 23 May 1997 (urban and peri-urban landscapes) and 17 April 1997 (agricultural landscape). We derived an isolation index for each patch as the distance to the nearest remnant > 10 ha in size. This form of patch isolation index has been found to be a useful predictor of 'effective isolation' in woodland birds studies in eastern Australia (Seddon *et al.*, 2001) and in other fragmented landscapes (e.g. van Dorp & Opdam, 1987; Delin & Andrén,

Table 2 Components of the avifaunal habitat complexity score (HCS), measured in 1 ha quadrats in woodland remnants (source: Watson, 1999; Freudenberger, 2001)

	HCS			
	0	1	2	3
Canopy layer*	0–10	10–20	20–50	> 50
Tall shrub layer†	0–10	10–20	20–50	> 50
Low shrub layer‡	0–10	10–20	20–50	> 50
Ground layer§	0–10	10–40	40–70	> 70
Logs and fallen branches¶	0–10	10–40	40–70	> 70
Litter layer**	0–10	10–40	40–70	> 70

Values are expressed as percentage.

*Estimated percentage foliage cover of trees > 4 m high.

†Estimated percentage foliage cover of shrubs 2–4 m high.

‡Estimated percentage foliage cover of shrubs 0.5–2 m high.

§Estimated percentage foliage cover of flora < 0.5 m high.

¶Estimated percentage of ground covered by logs and fallen branches.

**Estimated percentage of ground covered by dead leaves, sticks and twigs.

1999). The range of patch area and isolation values for each landscape is presented in Table 1 and illustrated in Figs 3 and 4.

Matrix characteristics

To quantify some ecologically relevant properties of each landscape, we estimated the tree cover and the amount of human disturbance in terms of houses and roads (Kluza *et al.*, 2000, Table 1). In detail, for each landscape we quantified the number of trees in paddocks, numbers of

houses, numbers and length of roads, numbers and length of vegetated creeklines, and the number of trees in clumps (i.e. more than one tree together) in 10 randomly placed circles of 1 km radius. We made sure that no circle contained any form of woodland remnant, i.e. they were placed in the matrix surrounding each woodland. All measurements were conducted using the SPOT imagery in the peri-urban and agricultural landscapes. Aerial photographs captured in May 2000 at a spatial resolution of 1 : 20,000 were used for the urban landscape. The SPOT imagery had a spatial resolution of 10 m, hence smaller trees and shrubs within the matrix were unlikely to be detected, but they were in any case rarely found during field surveys.

Analysis

Species-level analyses were restricted to species that are resident and breed only in woodland habitat because summer migratory species were leaving the study area and winter migratory species were arriving during the early autumn surveys. These species were defined *a priori* 'resident woodland birds' following the classifications of Taylor & Canberra Ornithologists Group (1992), Robinson & Traill (1996) and Watson *et al.* (2003). Those birds that occurred in fewer than five woodland remnants in any landscape were also excluded from the analyses because the logistic regression modelling technique could not pick up any area or isolation pattern for these species statistically. All statistics were calculated using SPSS version 11.5 (Kinnear & Gray, 2000).

Resident woodland bird species were further categorized by their body mass, foraging niche (insectivore, nectivore or granivore) and the foraging substrates they exploit (bark, ground, foliage, canopy or air) in order to examine the effects of the different matrices on functional groups of avifauna. Bird species were classified into these groups using data in Baker *et al.* (1997) (body mass) and Garnett *et al.* (1992) (foraging guilds and substrates).

As the matrix data captured were multivariate, a principal components analysis (PCA) was used to analyse and highlight the difference in the matrix characteristics of the three different landscapes. The seven matrix measures described in Table 1 were entered into the PCA as environmental variables. The PCA was conducted in Community Analysis Package software (Pisces Conservation, 2002).

Linear regressions of woodland-dependent species richness vs. remnant area and remnant isolation for the three different landscapes were calculated to test whether there were differences between them. Linear regressions of log-species on log-area and log-species on log-isolation for the three landscapes were plotted and compared for accuracy of fit to untransformed or semi-log plots. Species-log area and species-isolation regressions fitted best. The slope of each regression in each landscape was thus compared, using the methodology described in Zar (1996: 292–294). This method uses a *t*-test in a fashion analogous to that of testing for differences between two populations. The test statistic is

$$t = \frac{b_1 - b_2}{Sb_1 - b_2} \quad (1)$$

where the variables b_1 and b_2 are the regression coefficients and $Sb_1 - b_2$ is the standard error of the difference between the regression coefficients.

As patterns of spatial autocorrelation in data can influence results in these types of regression analysis, we conducted a spatial autocorrelation analysis in all three landscapes using SAAP 4.3 software (Wartenberg, 1989; Diniz-Filho *et al.*, 2003; Hawkins & Pausas, 2004). To evaluate the adequacy of the landscape regression models we generated spatial correlograms at eight distance classes of the raw woodland species richness data in all three landscapes, using Moran's *I* coefficients (Legendre, 1993).

A three-step procedure was used to determine if individual species had significant relationships between remnant area, isolation or both (i.e. Fig. 1), and if these relationships varied between the three landscapes. First, presence/absence data for all resident woodland-dependent species were modelled statistically using binomial logistic regressions to determine the relationships between the occurrence of each species and the size and isolation of woodland habitat, in each landscape. Secondly, if a species was found to have a significant ($P < 0.05$) relationship with area or isolation, a logistic regression equation was calculated to determine 'threshold' area and isolation sensitivities at an arbitrary 50% probability of occurrence, following the procedure and rationale of Åberg *et al.* (1995) and Helzer & Jelinski (1999). If both remnant area and isolation were found to be significant in explaining the presence/absence of a species in a landscape, a step-wise logistic regression model (following Åberg *et al.*, 1995) was used to determine the strength of each variable in explaining the species distribution. A new variable in the step-wise analysis was created to distinguish between compensatory effects and the block pattern (Fig. 1; following Lomolino, 1986). This variable equalled '1' when the habitat coordinate was within the block formed by the area and isolation minimum requirements and '0' when it was outside the region. This new 'block' variable was used in a step-wise regression analysis with area and isolation to test whether the 'block' variable best accounted for the observed pattern. If the block variable was a better fit than the other two variables (i.e. area and isolation), we assumed the species had a block pattern but if the former variable had a better fit, we assumed the species had a compensatory effect pattern. This technique was used to determine each species' pattern of distribution across the three landscapes.

RESULTS

Patch and landscape characteristics

Each landscape had similar remnant area and isolation ranges, means and medians, suggesting that remnant distribution in the three landscapes was comparable (in all cases, well within an order of magnitude, Table 1). Remnant area and remnant

isolation were independent of each other in all three landscapes (Fig. 3). There was a significant linear relationship between the log of remnant patch area and habitat complexity (condition) of the patch in all three landscapes (Fig. 4), indicating that small remnant patches, regardless of landscape, tended to have simpler habitat structure and larger remnant patches had more complex habitat structure.

The characteristics of the three landscape matrices were quite different. The agricultural matrix had fewer trees and nearly five times greater length of vegetated creeklines than the peri-urban matrix (Table 1). The amount of human settlement within the two matrices was also different with almost double the number of houses and triple the number of roads within the peri-urban landscape than the agricultural matrix. The urban matrix had a far greater number of houses, trees and road density than the other two landscapes. Two PCs accounted for 84.2% of the variation in matrix structure between the three landscapes. The first component separated the surveyed matrix circles on both the density (per km²) of man-made objects (e.g. houses and roads) and the density of trees and clumps of trees (Fig. 5). The second component separated the surveyed matrix circles on the density of creeks and length of creek lines within the matrix sample. This ordination clearly showed that there were differences in matrix composition among the three different landscapes. Predictably, there was overlap between the peri-urban and agricultural landscapes and no overlap between these landscapes and the urban landscape.

Avifauna of the region

In total, 130 bird species were identified within woodland remnants in the three landscapes: 115 in the agricultural

landscape, 95 in the peri-urban landscape and 90 in the Canberra urban landscape. The three landscapes had proportionally similar numbers of resident woodland species, with 54 species recorded from woodlands in the agricultural landscape, and the peri-urban and urban landscapes each having 43 species. Eleven woodland-dependent species were found only in the agricultural landscape (Black-chinned Honeyeater *Melithreptus gularis* Gould, Chestnut-rumped Heathwren *Hylacola pyrrhopygia* Mathews, Common Bronze-wing *Phaps chalcoptera* Latham, Grey-crowned Babbler *Pomatostomus temporalis* Vigors & Horsfield, Little Friarbird *Philemon citreogularis* Gould, Painted Button-quail *Turnix varia* Mathews, Peaceful Dove *Geopelia striata* Rand, Red-capped Robin *Petroica goodenovii* Mathews, Superb Parrot *Polytelis swainsonii* Mathews, White-browed Woodswallow *Artamus superciliosus* Gould and Striped Honeyeater *Plectorhyncha lanceolata* Gould). This may, in part, be due to the greater sampling effort ($n = 55$ woodlands) in the agricultural landscape compared with the peri-urban ($n = 42$) and the urban ($n = 30$) landscapes. Twenty-seven woodland dependent species were found in more than five remnants in all three landscapes (Table 3). These species were used to develop individual logistic regression models of occurrence as a function of remnant area and isolation.

Species richness

For each landscape, we found no significant spatial autocorrelation between woodland remnant species richness and the location of the remnants. There was a significant ($P < 0.0001$) relationship between resident woodland species richness and log remnant area in each of the three

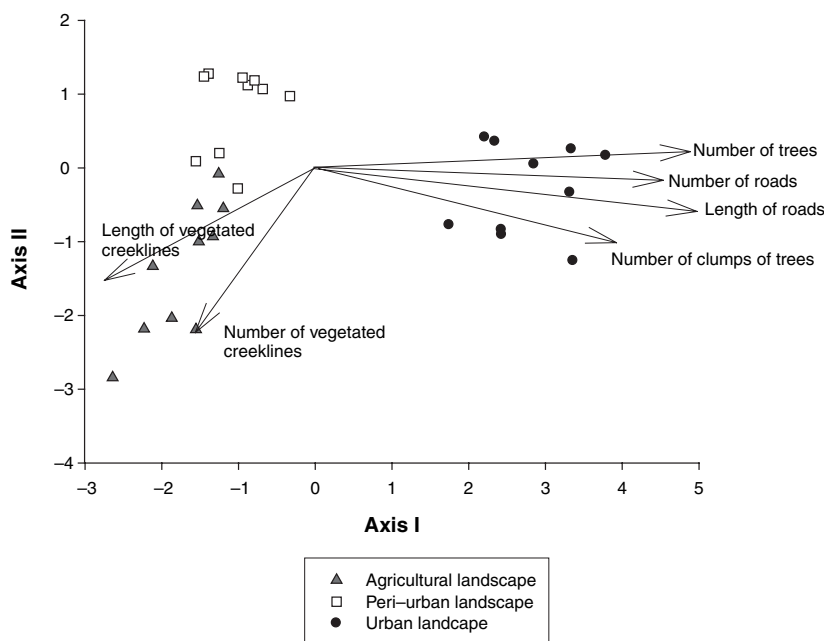


Figure 5 Ordination diagram of the first two axes of principal components analysis for matrix environmental variables in three fragmented woodland landscapes in south-eastern Australia. Axes I and II accounted for 62% and 22% of the variance in the environmental data. Arrows represent directions of greatest change of environmental variables. See Table 1 for definitions of environmental variables and how matrix variables were measured in the three landscapes.

Table 3 Mean deviance in individual-area and individual-isolation logistic regression models for resident woodland bird species in woodland remnants of three distinct landscapes – agricultural ($n = 55$), peri-urban ($n = 42$) and urban ($n = 30$) – in the region of the northern Australian Capital Territory and surrounding areas of New South Wales. Birds have been categorized into seven groups, based on their sensitivity to remnant area and remnant isolation

Bird name and category	Body size	Foraging substrate	Foraging habit	Agricultural		Peri-urban		Urban	
				Area	Isolation	Area	Isolation	Area	Isolation
Non-sensitive species									
Crimson Rosella <i>Platycercus elegans</i>	53	Canopy	Granivore	0.7	2.9	0.3	2.1	2.7	1.2
Striated Pardalote <i>Pardalotus striatus</i>	11	Shrub	Insectivore	0.3	0.4	0.4	1.4	0.7	0.2
Yellow-rumped Thornbill <i>Acanthiza chrysorrhoa</i>	9	Shrub	Insectivore	1.3	0.2	0.3	0.6	2.2	1.7
Restless Flycatcher <i>Myiagra inquieta</i>	20	Sallying	Insectivore	0.3	1.5	0.2	0.7	1.3	0.2
Mistletoebird <i>Dicaeum hirundinaceum</i>	9	Canopy	Nectivore	1.1	0.3	0.9	1.2	2.1	0.9
Noisy Friarbird <i>Philemon corniculatus</i>	115	Canopy	Nectivore	2.8	0.2	1.2	0.9	1.3	0.3
Noisy Miner <i>Manorina melanocephala</i>	58	Canopy	Insectivore	1.5	1.2	0.4	0.5	1.1	0.1
Dusky Woodswallow <i>Artamus cyanopterus</i>	35	Aerial	Insectivore	0.1	0.8	0.3	0.6	1.2	0.9
Sensitive to remnant area in all landscapes									
Brown Treecreeper <i>Climacteris picumnus</i>	33	Terrestrial	Insectivore	5.5**	2.6	7.5**	2.1	3.3*	2.3
Speckled Warbler <i>Chthonicola sagittata</i>	14	Terrestrial	Insectivore	4.3*	1.7	4.5*	1.1	4.8*	0.3
Striated Thornbill <i>Acanthiza nana</i>	6	Shrub	Insectivore	5.5*	3.1	3.2*	1.2	3.4*	0.24
Brown-headed Honeyeater <i>Melithreptus brevirostris</i>	14	Canopy	Nectivore	6.8**	1.5	4.3*	1.3	5.1**	0.7
Sensitive to remnant area and isolation only in the agricultural landscape									
White-throated Treecreeper <i>Cormobates leucophaeus</i>	22	Bark	Insectivore	3.5*	9.6**	8.3**	1.3	3.3**	0.6
Buff-rumped Thornbill <i>Acanthiza reguloides</i>	8	Shrub	Insectivore	6.8*	6.7**	10.1**	2.1	7.3**	2.3
Eastern Spinebill <i>Acanthorhynchus tenuirostris</i>	11	Shrub	Nectivore	3.9*	3.8*	4.1*	0.7	4.2*	0.5
Little Wattlebird <i>Anthochaera carunculata</i>	110	Canopy	Nectivore	9.6**	7.1**	4.1**	2.9	5.6**	0.4
Eastern Yellow Robin <i>Eopsaltria australis</i>	19	Ground	Insectivore	9.3**	8.1**	4.1*	1.6	5.1**	0.4
White-winged Chough <i>Corcorax melanorhamphos</i>	358	Terrestrial	Insectivore	3.5*	6.8**	7.7**	2.9	8.8**	0.1
Sensitive to isolation only in the agricultural landscape									
Striated Thornbill <i>Acanthiza lineata</i>	7	Shrub	Insectivore	0.2	5.6**	1.1	2.3	0.5	1.4
White-eared Honeyeater <i>Lichenostomus leucotis</i>	20	Shrub	Insectivore	1.8	6.4**	2.1	1.8	0.9	0.8
Fuscous Honeyeater <i>Lichenostomus fuscus</i>	17	Canopy	Nectivore	3.1	4.7*	2.2	2.3	0.1	1.5
Sensitive to area only in the urban landscape									
Crested Pigeon <i>Ocyphaps lophotes</i>	210	Terrestrial	Granivore	0.2	0.3	1.1	0.1	5.9**	0.4
Eastern Rosella <i>Platycercus eximius</i>	99	Canopy	Nectivore	1.2	2.7	0.9	1.5	7.4**	3.1
Jacky Winter <i>Microeca fascians</i>	14	Terrestrial	Insectivore	0.2	1.7	1.4	0.2	3.8*	0.1
Brown Shrike-thrush <i>Colluricincla harmonica</i>	63	Shrub	Insectivore	0.5	0.1	2.3	2.1	8.5**	0.1
Sensitive to isolation in the agricultural landscape and area in peri-urban and agricultural landscapes									
Brown Thornbill <i>Acanthiza pusilla</i>	9	Shrub	Insectivore	0.9	6.4**	6.7**	0.3	4.1*	0.9
Sensitive to area in peri-urban and urban landscapes									
Weebill <i>Smicrornis brevirostris</i>	6	Shrub	Insectivore	0.9	0.5	8.4**	0.2	3.4*	0.1

Data for body mass, foraging niche (insectivore, nectivore and granivore) and foraging substrates (bark, ground, foliage, canopy and air) from Baker *et al.* (1997) (body mass) and Garnett *et al.* (1992) (foraging guilds and substrates).

* $P < 0.05$, ** $P < 0.01$.

landscapes (Fig. 6). The tightest species–area relationship, with the greatest slope, was found in the urban landscape. The slopes of the urban and peri-urban regressions ($t = 2.42$, d.f. = 68, $P < 0.005$) and the urban and agricultural regressions ($t = 2.95$, d.f. = 80, $P < 0.0025$) were

significantly different, but the slopes of the peri-urban and agricultural regressions ($t = 0.87$, d.f. = 92, $P > 0.10$) were not. When remnant isolation was considered, we found that there was a significant ($P < 0.01$) relationship between remnant isolation and resident woodland species richness

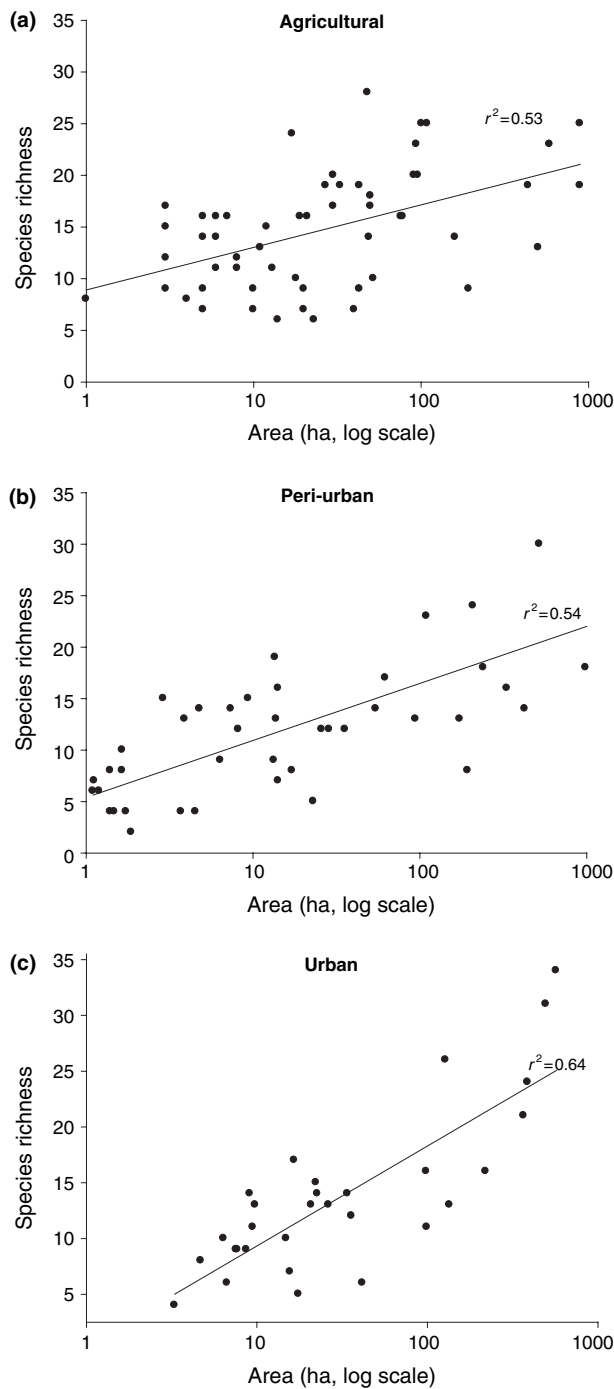


Figure 6 Woodland bird species richness (SP) vs. log area (A) regressions for woodland patches in (a) agricultural landscape, (b) peri-urban landscape and (c) urban landscapes [urban: $F_{1,28} = 50.4$, $P < 0.0001$, $r^2 = 0.64$, $SP = 0.37 + 8.95 \log(A)$; peri-urban: $F_{1,40} = 46.4$, $P < 0.0001$, $r^2 = 0.53$, $SP = 5.3 + 5.5 \log(A)$; agricultural: $F_{1,53} = 21.7$, $P < 0.0001$, $r^2 = 0.54$, $SP = 8.9 + 4.11 \log(A)$].

only in the agricultural landscape (Fig. 7). These results suggest that the urban matrix has a different and greater effect on the species–area relationship than the two other

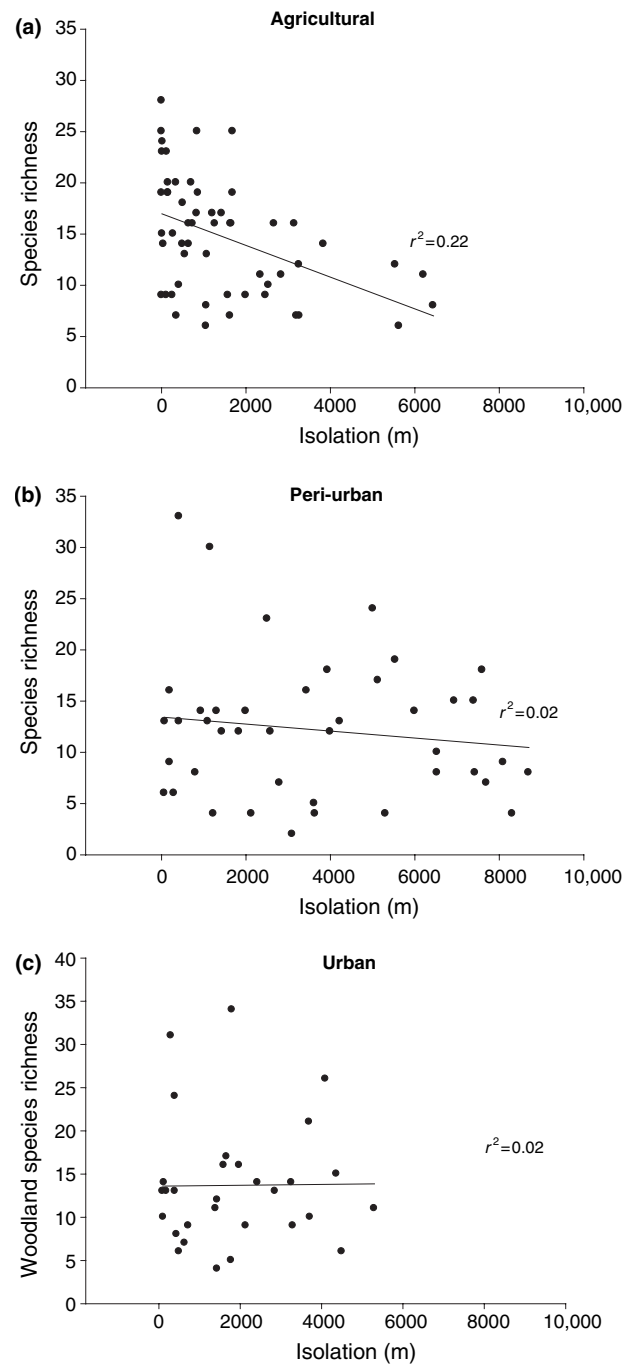


Figure 7 Woodland bird species richness (SP) vs. isolation (I) regressions for woodland patches in (a) agricultural landscape, (b) peri-urban landscape and (c) urban landscapes. Remnant isolation was calculated as the distance to the nearest remnant > 10 ha in size. Only the agricultural landscape species–isolation regression was significant [agricultural: $F_{1,53} = 15.1$, $P < 0.001$, $r^2 = 0.22$, $SP = 16.97 - 1.54(I)$; urban: $F_{1,28} = 0.03$, $P = 0.95$, $r^2 = 0.01$; peri-urban $F_{1,40} = 0.76$, $P = 0.38$, $r^2 = 0.02$].

matrices and that the agricultural matrix appears to have an influence on species sensitivity to isolation, whereas the other two matrices do not.

Individual species responses

Of the 27 species analysed, remnant area explained a significant ($P < 0.05$) portion of the deviance in the presence/absence logistic models of 16 resident woodland species, in one or more landscape (Table 3). Ten species were found to be area-sensitive in all three landscapes, while two species were area-sensitive in the agricultural and peri-urban landscapes, and four species were found to be area-sensitive only in the urban landscape (Table 3). Remnant isolation explained a significant ($P < 0.05$) portion of the deviance in the presence/absence logistic models of 10 woodland-resident species, but only in the agricultural system (Table 3). Eight resident woodland species were affected neither by remnant area nor isolation in any of the landscapes.

Using the results of the logistic analyses, we identified six broad patterns of sensitivity to remnant area and/or isolation in the three landscapes (Table 3). The largest group ($n = 8$) was of species considered 'tolerant', as they occurred in most or all remnant sizes, irrespective of the degree of isolation. A second group of four species was found to be area-sensitive in all three landscapes, but not sensitive to isolation in any landscape. A further four species were area-sensitive only in the urban landscape and not sensitive to isolation in any landscape. Two species were found to be area-sensitive in peri-urban and agricultural landscapes but not in urban landscapes, with one of these species (Brown Thornbill *Acanthiza pusilla*) appearing to be sensitive to isolation in the agricultural landscapes.

Six species were significantly ($P < 0.01$) affected by remnant area and isolation in the agricultural landscape (Table 3). All six species were found to have a compensatory effect between area and isolation (e.g. Eastern Yellow Robin *Eopsaltria australis* Shaw; Fig. 8) and no species displayed a 'block' pattern (Fig. 1). Another group comprising three species were sensitive only to isolation in the agricultural landscape and were not area-sensitive in any landscape (Table 3).

The 'threshold' values, defined as the area/isolation where 50% probability of occurrence is reached, for area-sensitive species ranged from 5 ha (Weebill *Smicrornis brevirostris* Gould in urban landscapes) to 990 ha (Eastern Yellow Robin *E. australis* in peri-urban landscapes). Many species found to be area-sensitive in all three landscapes were found to have markedly different threshold values in each landscape (Table 4). For example, the Brown Treecreeper *Climacteris picumnus* Temminck had a threshold of 40 ha in the agricultural landscape, 320 ha in the urban landscape and 410 ha in the peri-urban landscape. The Eastern Yellow Robin *E. australis* was equally variable in its threshold: 73 ha in the agricultural landscape, 101 ha in the urban landscape and 990 ha in the peri-urban landscape (Table 4).

Woodland-dependent birds were significantly affected by remnant isolation only in the agricultural landscape. The isolation threshold values (50% probability of occurrence) varied from 200 to 2600 m between species, with the

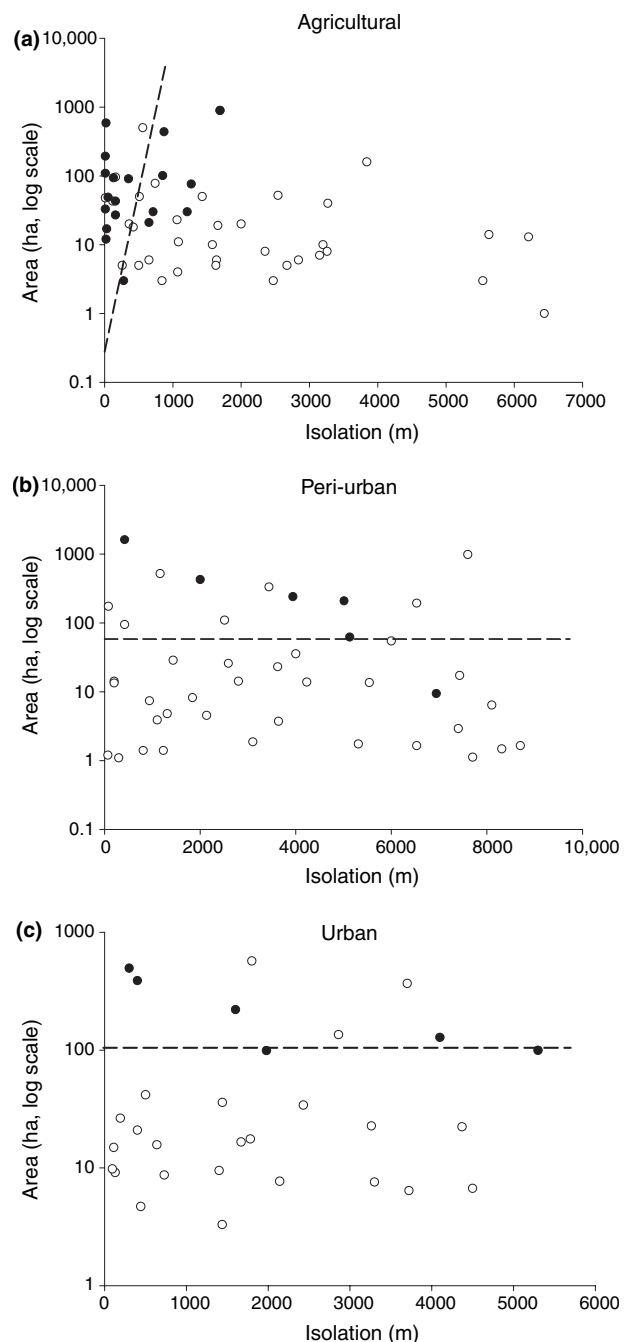


Figure 8 Area-isolation plots for Eastern Yellow Robin *Eopsaltria australis* in three fragmented woodland landscapes – (a) agricultural, (b) peri-urban and (c) urban – in south-eastern Australia. Open circles are unoccupied remnants, while closed circles represent remnants in which the species occurs. This species displayed a minimum area pattern in urban and peri-urban landscapes and a compensatory pattern in the agricultural landscape (see Fig. 1). Dashed lines represent the arbitrary species threshold where above the filled line the probability of occurrence is $> 50\%$. Thresholds were determined using a logistic regression equation, following Åberg *et al.* (1995) and Helzer & Jelinski (1999).

Table 4 'Threshold' area (ha) and isolation (m) values of bird species in woodland remnants of three landscapes (agricultural, peri-urban and urban). Following Helzer & Jelinski (1999), threshold values were defined as the area or the isolation measure where a species reached 50% probability of occurrence in their individual area and isolation incidence functions. Incidence functions were developed from logistic regressions based on the presence/absence data for bird species in each landscape. If a species did not reach 50% probability of occurrence within the limits of the data, no minimum area requirement was calculated

Bird name and category	Agricultural		Peri-urban		Urban	
	Area	Isolation	Area	Isolation	Area	Isolation
Sensitive to remnant area						
<i>Climacteris picumnus</i>	40	–	410	–	320	–
<i>Chthonicola sagittata</i>	390	–	365	–	180	–
<i>Acanthiza nana</i>	40	–	30	–	35	–
<i>Melithreptus brevirostris</i>	160	–	40	–	45	–
Sensitive to remnant area (all landscapes) and to isolation in agricultural landscapes						
<i>Cormobates leucophaeus</i>	3	2600	19	–	60	–
<i>Acanthiza reguloides</i>	18	1640	13	–	41	–
<i>Acanthorhynchus tenuirostris</i>	580	450	310	–	410	–
<i>Anthochaera carunculata</i>	40	780	35	–	45	–
<i>Eopsaltria australis</i>	73	560	990	–	101	–
<i>Corcorax melanorhamphos</i>	190	260	209	–	105	–
Sensitive to isolation in the agricultural landscape						
<i>Acanthiza lineate</i>	–	1270	–	–	–	–
<i>Lichenostomus leucotis</i>	–	200	–	–	–	–
<i>Lichenostomus fuscus</i>	–	230	–	–	–	–
Sensitive to area in the urban landscape						
<i>Microeca fascians</i>	–	–	–	–	140	–
<i>Colluricincla harmonica</i>	–	–	–	–	200	–
<i>Ocyphaps lophotes</i>	–	–	–	–	40	–
<i>Platycercus eximius</i>	–	–	–	–	35	–
Sensitive to isolation in agricultural landscapes and area in peri-urban and rural landscapes						
<i>Acanthiza pusilla</i>	–	–	35	–	140	–
Sensitive to area in peri-urban and rural landscapes						
<i>Smicrornis brevirostris</i>	–	–	90	–	5	–

White-throated Treecreeper *Cormobates leucophaeus* Latham having the largest threshold distance (Table 4).

DISCUSSION

Whilst this study comprises a large sample effort, with 127 woodlands enumerated, we have no replication of landscapes. We therefore cannot claim that the relationships described herein are characteristic of each landscape type, e.g. that the species–incidence relationships reported for Canberra would be repeated for other urban landscapes in eastern Australia. However, we have shown that resident woodland bird species differed in their response to remnant patch size and isolation across the three contrasting landscapes. Whilst it is important to acknowledge that there might be ecologically relevant differences in the properties of the woodland remnants that were not quantified in our study, it appears unlikely that intrinsic properties of the remnants could account for the

pronounced differences in species incidences. This is because the spatial pattern, size distribution, level of disturbance to habitat and floristic composition of woodland remnants were similar in all three landscapes (Figs 3 and 4; Table 1; Watson, 1999; Freudenberger, 2001). Moreover, each landscape experiences a similar climate due to its proximity to each other and similar altitude (Bureau of Meteorology, 2004). We therefore consider that the variations in species relationships with remnant area and isolation are most parsimoniously explained as a function of differences in the properties of the landscape matrix surrounding the woodland remnants.

It was beyond the scope of this study to discern the reasons why species had varying area-sensitivity and isolation-sensitivity responses in these landscapes, but there are a number of plausible hypotheses for how the condition of the matrix could affect bird distribution within woodland remnants. First, different matrix conditions will affect the 'edge-contrast' between woodland remnant and matrix, which

will change many edge and internal properties of a woodland fragment. This is because the 'edge-contrast' governs the micro-climate gradient at the edge of a remnant, which in turn could lead to substantial vegetation changes (including changes in species composition and structure) and biological changes (including increases in predators, nest predators and nest parasites and changes in food resources) along the new edge (Robinson *et al.*, 1995; Kapos *et al.*, 1997; Zannette *et al.*, 2000; Rankin-De Mérona & Hutchings, 2001). This could change the rates of nest predation, nest parasitism or inter-specific competition within woodland remnants, as well as the availability of food and resources, all of which will affect a bird species' area-sensitivity (Ambuel & Temple, 1983; Wilcove, 1985; Soulé *et al.*, 1988; Robinson, 1992; Burke & Nol, 1998; Grey *et al.*, 1998; Crooks & Soule, 1999). Secondly, different landscape matrices have different resources within them, and therefore have differential capacity for different species – if a species cannot use the resources of a particular matrix, the population may be dependent on a bigger area within the remnant habitat itself to sustain a population in a fragment (Lindenmayer *et al.*, 2003).

This variation in individual species responses underpins the differences in species–area relationships between the three landscapes (Fig. 6). Wethered & Lawes (2003) report a similar matrix effect for birds in the Afromontane Mistbelt of South Africa, as they found a strong species–area effect in forest patches surrounded by grassland but no effect for forest patches surrounded by commercial plantation matrix. They concluded that commercial plantations may increase the likelihood of movements of birds into small native forest patches. We found that the urban landscape had the steepest slope (i.e. had the greatest area effect). In addition to threats mentioned above, bird populations in small urban woodland remnants may be threatened by factors such as (1) greater competition with exotics such as European Starlings *Sturnus vulgaris* Linnaeus (Kerpez & Smith, 1990), (2) exposure to larger populations of predators such as domestic cats (Marzluff & Restani, 1999), and (3) heightened disturbance and mortality from human activity (Knight & Gutzwiller, 1995; Evans, 1998). However, it is not the case that the urban landscape was the most 'hostile' (sensu Watson, 2002) matrix for all species. Rather, trends in area-sensitivity were shown to be species-specific and not landscape-specific (Tables 3 and 4).

Woodland-dependent species richness was significantly ($P < 0.01$) correlated with remnant isolation only in the agricultural landscape (Fig. 7), within which we found 10 individual species that were statistically sensitive to remnant isolation (Table 3). This may be due to a lack of connectivity within the agricultural landscape matrix as there are fewer paddock trees within the matrix and therefore the remnants are acting more like true islands. Alternatively, or additionally, the agricultural matrix may be home to different alien species and threatening processes (e.g. predators), which could impact the dispersal capabilities of some species between remnants (Andrén, 1994; Stouffer & Bierregaard, 1995b; Sisk *et al.*, 1997). Interestingly, we found that seven species were sensitive

to area and isolation in the agricultural landscapes, with all species displaying a compensatory effect (Fig. 1e). That is, these species occurred in small remnants only if they were weakly isolated. This pattern has been described in insular mammals in the Thousand Islands region of New York (Lomolino, 1986) but was not found in resident bird communities in humid pine-oak forest fragments in Oaxaca, southern Mexico (Watson, 2002).

The role of the matrix in conservation biology and ecological theory

Numerous approaches in conservation biology, from theoretical modelling (e.g. Lindenmayer & Lacy, 1995; With, 2002) to practically oriented ecology (e.g. Lambeck, 1997; Brooker, 2002) have been used to explore and explain how species react to habitat loss and fragmentation. Many of these models or approaches ignore matrix characteristics as an explanatory factor. The results in this study suggest that there is a danger of over-generalizing and thus trivializing species–area and species–isolation relationships. Our results indicate that the structure, composition and disturbance regimes of the matrix may strongly influence both the area-sensitivity and/or dispersal behaviour of a bird species in highly fragmented landscapes. If the matrix properties are not considered in metapopulation modelling, for example, the projections of the survival probability may be incorrect, as these values may not necessarily predict the immigration rates into patches (Tischendorf *et al.*, 2003). Landscape matrices cannot be assumed to be a hostile 'sea' of non-habitat. For some species the matrix may be habitat most of the time (Manning *et al.*, 2004), while for other species a 50-m gap between patches of forests may be a sufficient barrier to dispersal (Brooker & Brooker, 1997, 2001).

Furthermore, the search for general biogeographical patterns such as body size and area–incidence relationships (e.g. Bierdermann, 2003), or general biogeographical theories that explain species presence in fragmented landscapes (e.g. Lomolino, 2000; Watson, 2002) may be problematic when landscape context is considered. For example, we showed that species with the same mass may have very different probability of occurrences (Table 4) in landscapes with different matrices surrounding remnants, thereby confounding the general pattern suggested by Bierdermann (2003). This finding goes against other studies that have shown that body mass is correlated with minimum patch size (e.g. Leck, 1979; Beier *et al.*, 2002) and is inconsistent with the hypothesis that if home range size scales to body mass, area-sensitive birds will be absent from patches smaller than a few home ranges (Stratford & Stouffer, 1999). In our study area, some small-bodied birds may have relatively higher minimum patch size requirements because they may be more susceptible to small environmental perturbations (Johst & Brandl, 1997; Gaston & Blackburn, 2000), which will occur with more frequency in small remnants.

Lomolino's (2000) species-based theory of insular zoogeography employs the concept of 'immigration filters' to show the impacts of the matrix on area–isolation relationships of

species. In our study we did not find a common response to matrix condition. Rather, species area vs. isolation relationships can change considerably across landscapes with markedly different matrices, but similarly fragmented remnants. While identifying broad general biogeographical patterns can be useful to guide conservation planning in fragmented landscapes, the assumption that species or functional groups will conform to remnant area reduction and remnant isolation in a consistent fashion, needs to be re-addressed on the basis of this empirical evidence. Overlooking processes such as differential responses to matrix conditions could lead to the creation of too broad a framework or theory, and perhaps ultimately poor conservation strategies (Lindenmayer & Franklin, 2002; Lomolino & Smith, 2003; MacDonald & Kirkpatrick, 2003).

Conservation implications

Perhaps the clearest finding in this study in terms of bird conservation in the region is the large number of resident woodland species (in particular, small insectivorous species) that were found to be area-sensitive in all landscapes. This area effect in resident woodland bird species is now well documented in Australia (e.g. Saunders, 1993; Zannette *et al.*, 2000; Major *et al.*, 2001; MacNally & Horrocks, 2002; Watson *et al.*, 2003). The majority of these studies argue that conservation strategies should protect larger woodland remnants, buffer and rehabilitate small woodland remnants and create habitat corridors. We agree that this is a logical starting point, but this and other recent work (e.g. Margules & Pressey, 2000; Fischer & Lindenmayer, 2002c) leads us to suggest that small habitat patches and appropriate matrix management can be valuable components of a conservation strategy. In fact at the scale involved in our system, corridors may not be required for many bird species if matrix conditions allow for adequate dispersal and colonization. Small patches, isolated trees or urban street trees in the matrix may serve as 'stepping stones' and thus facilitate movement through the landscape; especially in the agricultural landscape where some bird species are affected by remnant isolation (Potter, 1990; Fischer & Lindenmayer, 2002a,b). Moreover, it is likely that small patches have other ecologically desirable functions –for example, they may stabilize soils, reduce salinity and provide habitat for less area-sensitive species like small reptiles and invertebrates (summarized in Fischer & Lindenmayer, 2002a). Therefore, our findings are consistent with research encouraging a more multifaceted conservation approach than typifies earlier work viewing woodland fragments as islands in an essentially hostile sea of non-habitat. We strongly encourage conservation programmes that maintain woodland vegetation within the matrix habitats, especially agricultural environments where paddock trees are rare and declining (Ozolin *et al.*, 2001; Gibbons & Boak, 2002) and that consider the protection of both small and large woodland remnants in severely fragmented landscapes.

CONCLUSION

Protecting biodiversity in fragmented landscapes will require substantial conservation and ecosystem restoration efforts (Young, 2000; Lindenmayer *et al.*, 2002). In the past, these efforts have focused on either creating corridors or buffering remnant areas to make them bigger (Marzluff & Ewing, 2001). Our study strongly suggests that landscape matrix characteristics play an important role in determining how individual species respond to processes associated with habitat fragmentation. Species cannot be assumed to respond in a consistent fashion across landscapes because fragmentation processes appear to interact with matrix conditions to produce highly individualistic responses. Indeed, it is noteworthy how substantial the variation in form of incidence functions and critical areas for patch occupation is for some of our study species. With this paper we hope to stimulate further research and discussion concerning the factors controlling such variation and how to develop approaches for landscape conservation in fragmented landscapes that incorporate matrix management. To assist conservation and restoration efforts, more research is needed on how the matrix affects animal distributions in fragmented landscapes; especially how different taxa disperse through different landscapes.

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